MAT 127A Discussion 10 Notes

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§I Uniform Continuity

A function $f : A \to \mathbb{R}$ is uniformly continuous on A if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all $x, y \in A$, $|x - y| < \delta$ implies $|f(x) - f(y)| < \varepsilon$.

> Remark: Compare this definition with continuity: f is continuous on A if for every $c \in A$ and $\varepsilon > 0$, there exists a $\delta > 0$ such that for all $x \in A$, $|x - c| < \delta$ implies $|f(x) - f(c)| < \varepsilon$. To prove continuity, the choice of δ can depend on both ε and the point c.
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> In the definition of uniform continuity, y takes the role of c, but it is introduced after δ which means that the choice of δ can only depend on ε.

> (Exercise 4.4.1) Let $f(x) = x^3$. Show that f is uniformly continuous on any bounded subset of $\mathbb{R}$.

Proof. Let $A \subseteq \mathbb{R}$ be bounded. Thus, there exists $M > 0$ such that for all $x \in A$, we have $|x| \leq M$. Let $\varepsilon > 0$. Note that for all $x, y \in A$, we have

$$|f(x) - f(y)| = \left|x^3 - y^3\right| = |x - y|\left|x^2 + xy + y^2\right| \leq 3M^2|x - y|.$$

Thus, if we set $\delta = \frac{\varepsilon}{3M^2}$, then for all $x, y \in A$, we have that $|x - y| < \delta$ implies

$$|f(x) - f(y)| \leq 3M^2|x - y| < 3M^2\delta = \varepsilon.$$

■

Theorem 4.4.5 relates sequences and the absence of uniform continuity: A function $f : A \to \mathbb{R}$ fails to be uniformly continuous on A if and only if there exist $\varepsilon_0 > 0$ and sequences $x_n, y_n \in A$ such that $|x_n - y_n| \to 0$ but $|f(x_n) - f(y_n)| \geq \varepsilon_0$ for all $n \in \mathbb{N}$.

This result is important because we have defined compactness in terms of sequences and have gave many concepts sequential characterizations (e.g., continuity, closed sets, etc.). When we want to study the relation between uniform continuity and those concepts, Theorem 4.4.5 will be very useful.

> 1. (Exercise 4.4.2) Show that $f(x) = \frac{1}{x}$ is not uniformly continuous on $(0, 1)$.
> 2. Show that $f(x) = x^2$ is not uniformly continuous on $\mathbb{R}$.

Proof.

1. Let $x_n = \frac{1}{n}$ and $y_n = \frac{1}{n+1}$ for all $n \in \mathbb{N}$, and fix $\varepsilon_0 = 1$. Then, we have

   $$|x_n - y_n| = \left|\frac{1}{n} - \frac{1}{n+1}\right| = \frac{1}{n(n+1)} \to 0,$$

   but

   $$|f(x_n) - f(y_n)| = |n - (n+1)| = 1 \geq \varepsilon_0.$$

   Thus, f is not uniformly continuous on $(0, 1)$.

2. Let $x_n = n$ and $y_n = n + \frac{1}{n}$ for all $n \in \mathbb{N}$, and fix $\varepsilon_0 = 2$. Then, we have

$$|x_n - y_n| = \left| n - \left(n + \frac{1}{n} \right) \right| = \frac{1}{n} \to 0,$$

but

$$|f(x_n) - f(y_n)| = \left| n^2 - \left(n + \frac{1}{n} \right)^2 \right| = \left| n^2 - \left(n^2 + 2 + \frac{1}{n^2} \right) \right| = 2 + \frac{1}{n^2} > \varepsilon_0.$$

Thus, f is not uniformly continuous on $\mathbb{R}$. ■

> 1. Give an example of a continuous function f on a bounded set A such that $f(A)$ is unbounded.
> 2. Show that if f is uniformly continuous on a bounded set A, then $f(A)$ is bounded.

Proof.

1. An example is $f(x) = \frac{1}{x}$ on the bounded set $A = (0, 1)$. We have $f(A) = (1, \infty)$.
2. *Proof 1.* (Using Theorem 4.4.5 and Bolzano-Weierstrass theorem) Suppose for the sake of contradiction that $f(A)$ is unbounded. Let $x_1 \in A$ be arbitrary (note that $f(A)$ being unbounded implies that A is nonempty). Recursively, after we fix $x_k \in A$, we can find $x_{k+1} \in A$ such that

$$|f(x_{k+1})| \geq |f(x_k)| + 1$$

because $f(A)$ is unbounded. Since $|f(x_{k+1})| - |f(x_k)| \geq 1$ for all $k \in \mathbb{N}$, we know that for all $m, n \in \mathbb{N}$ with $m > n$, we have

$$|f(x_m) - f(x_n)| \geq |f(x_m)| - |f(x_n)| \geq m - n \geq 1.$$

Now, since A is bounded, by the Bolzano-Weierstrass theorem, there exists a subsequence (x_{n_k}) of (x_n) converging to some limit point $c \in \mathbb{R}$. Thus, we have

$$0 \leq \left| x_{n_{k+1}} - x_{n_k} \right| \leq \left| x_{n_{k+1}} - c \right| + \left| x_{n_k} - c \right|,$$

which implies that $\left| x_{n_{k+1}} - x_{n_k} \right| \to 0$ by the squeeze theorem. However, we have

$$\left| f\left(x_{n_{k+1}}\right) - f\left(x_{n_k}\right) \right| \geq 1$$

This contradicts with the fact that f is uniformly continuous on A, thus, $f(A)$ must be bounded.

Proof 2. (Using the definition of uniform continuity) Let $\varepsilon = 1$. Since f is uniformly continuous on A, there exists $\delta > 0$ such that for all $x, y \in A$, if $|x - y| < \delta$, then $|f(x) - f(y)| < 1$. Since A is bounded, there exists $M > 0$ such that $A \subseteq [-M, M]$. We can find a finite cover $[-M, M]$ by nonempty intervals of length less than δ, say, $(a_1, b_1), (a_2, b_2), \ldots, (a_N, b_N)$. Thus, we have $A \subseteq \cup_{i=1}^N (a_i, b_i) \cap A$. Now, uniform continuity implies that for each i and every two points $y_1, y_2 \in f((a_i, b_i) \cap A)$, we have $|y_1 - y_2| < 1$. In particular, we have

$$f((a_i, b_i) \cap A) \subseteq V_1\left(f\left(\frac{a_i + b_i}{2} \right) \right).$$

Thus, we have

$$f(A) \subseteq \cup_{i=1}^N f((a_i, b_i) \cap A) \subseteq \cup_{i=1}^N V_1\left(f\left(\frac{a_i + b_i}{2} \right) \right),$$

which is bounded because finite unions of bounded sets is bounded. ■